

TOORA-KHEM, Russian Federation

WMO: 361030

Lat: 52.4703N

Lon: 96.1192E

Elev: 3017

StdP: 13.16

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.4	-37.8	-45.8	0.4	-41.6	-41.7	0.6	-37.5	12.8	3.1	9.8	1.2	0.1	340	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.1	82.8	62.1	79.0	60.3	75.5	59.6	64.6	76.6	62.9	74.5	61.2	72.3	4.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.4	87.8	68.9	58.6	82.2	67.1	57.1	77.7	65.7	31.4	76.1	30.0	74.2	28.8	72.7	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.8	10.1	8.3	-48.4	88.4	5.4	4.0	-52.3	91.3	-55.5	93.7	-58.6	95.9	-62.5	98.9		
(5)			-47.1	68.7	4.2	3.0	-50.1	70.8	-52.6	72.6	-55.0	74.3	-58.0	76.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.0	-13.1	-1.8	16.9	33.4	44.2	55.7	60.0	56.6	44.1	28.4	5.4	-7.7		
	DBStd	26.88	9.98	10.87	11.26	8.34	7.36	7.41	5.15	5.68	7.31	9.47	13.14	11.90		
	HDD50	9185	1955	1449	1026	500	211	30	3	11	203	670	1339	1788		
	HDD65	13895	2420	1869	1491	949	644	286	166	265	628	1135	1789	2253		
	CDD50	789	0	0	0	2	32	202	312	215	26	0	0	0		
	CDD65	24	0	0	0	0	0	8	11	5	0	0	0	0		
	CDH74	1077	0	0	0	0	70	417	380	201	9	0	0	0		
	CDH80	249	0	0	0	0	10	109	89	41	0	0	0	0		
	WSAvg	2.4	0.9	1.4	2.5	3.7	4.1	3.2	2.5	2.4	2.5	2.3	1.8	1.1		
	PrecAvg	13.1	0.4	0.2	0.2	0.6	1.2	2.0	2.7	2.7	1.5	0.6	0.6	0.6		
(15) Precipitation	PrecMax	19.0	0.7	0.6	0.6	1.1	2.5	3.5	5.3	4.8	3.3	1.5	1.2	0.9		
	PrecMin	8.0	0.1	0.0	0.0	0.1	0.2	0.7	0.7	0.8	0.4	0.1	0.2	0.2		
	PrecStd	2.6	0.2	0.1	0.1	0.2	0.6	0.7	1.1	1.0	0.7	0.3	0.3	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.5	33.0	52.3	68.7	80.7	87.8	87.7	84.7	74.6	58.9	36.5	26.2		
		MCWB	16.8	27.7	40.0	48.6	56.0	63.4	65.2	63.7	57.2	46.3	31.7	24.9		
	2%	DB	12.6	26.7	46.5	61.7	75.2	83.4	82.0	79.1	69.3	52.4	31.9	20.3		
		MCWB	11.4	23.8	36.1	44.9	53.5	61.4	62.4	62.3	54.7	43.1	29.4	19.0		
	5%	DB	7.1	21.2	41.6	56.0	69.4	78.9	78.1	75.2	64.3	46.6	28.6	15.6		
		MCWB	6.1	18.3	33.6	42.4	50.6	59.5	61.3	61.2	52.9	39.4	25.9	14.5		
	10%	DB	3.0	15.9	36.2	50.4	63.7	74.8	74.2	70.9	59.2	41.7	24.8	10.8		
		MCWB	2.1	13.6	30.0	39.8	48.2	58.2	60.8	58.5	49.8	36.5	22.6	9.9		
	0.4%	WB	17.0	28.5	40.1	50.1	58.0	66.3	67.2	66.3	58.8	47.5	32.6	25.0		
		MCDB	18.3	32.5	51.0	64.3	78.8	82.1	81.9	79.5	71.0	55.9	34.4	26.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	11.3	23.7	36.8	45.9	54.7	63.4	64.8	63.9	56.2	43.5	29.5	19.1		
		MCDB	12.5	26.9	45.2	59.0	72.5	78.3	75.9	75.8	67.1	51.5	31.9	20.3		
	5%	WB	6.2	18.5	33.6	43.1	51.8	61.2	63.3	61.8	53.4	39.9	26.1	14.5		
		MCDB	7.1	21.2	40.8	54.3	67.2	75.2	73.8	72.5	62.8	45.9	28.4	15.6		
	10%	WB	2.1	13.7	30.3	40.2	49.0	59.2	61.8	59.8	50.4	36.6	22.5	9.9		
		MCDB	3.0	15.7	36.2	50.4	61.9	71.8	72.1	68.9	58.2	41.1	24.8	10.9		
	5% DB	MDBR	26.8	32.7	33.0	26.1	27.9	28.3	25.1	24.8	26.6	23.5	24.1	22.9		
		MCDBR	25.7	31.3	35.2	36.9	42.5	39.2	35.3	35.5	37.2	30.6	22.1	21.4		
		MCWBR	24.3	27.2	25.1	22.0	23.5	19.2	17.3	19.4	22.3	20.6	18.5	20.0		
		MCWBR	25.3	30.4	32.3	33.6	38.3	32.4	26.9	30.5	34.7	29.2	20.4	21.0		
(35) Mean Daily Temperature Range	5% WB	MCWBR	24.0	26.4	23.1	20.6	21.8	16.1	14.3	17.0	20.8	20.0	17.1	19.7		
(40) Clear-Sky Solar Irradiance	taub		0.257	0.287	0.329	0.364	0.377	0.370	0.373	0.359	0.325	0.311	0.289	0.261		
	taud		2.137	2.106	2.087	2.244	2.286	2.412	2.429	2.434	2.478	2.410	2.207	2.090		
	Ebn at Noon		231	258	269	275	277	280	277	275	273	252	215	201		
	Edh at Noon		22	31	40	39	39	35	34	32	27	23	21	19		
(44) All-Sky Solar Radiation	RadAvg		403	739	1226	1573	1699	1887	1739	1475	1077	673	376	285		
	RadStd		23	40	49	76	134	156	96	106	126	51	36	19		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page